

Set 2 Answer

Scenario based

1. Retail Sales Forecasting

Logic

1. Load the retail sales dataset.
2. Clean and preprocess the data.
3. Separate input features and target (Sales).
4. Apply feature selection (RFE or SelectKBest) to identify the most important features.
5. Select the best features.
6. Train the forecasting model.
7. Predict future product sales.

2. Podcast Recommendation

Logic

1. Collect podcast details such as category and host.
2. Collect the user's listening history.
3. Create a user profile based on preferred categories and hosts.
4. Compare available podcasts with the user's interests.
5. Calculate similarity between user preferences and podcast features.
6. Recommend podcasts with the highest similarity scores.

3. Machine Failure Prediction

Logic

1. Load the machine sensor dataset.
2. Clean missing values.
3. Separate features and target (Machine Failure).
4. Standardize the data if required.
5. Apply PCA to reduce dimensionality or RFE to select important features.
6. Train the prediction model.
7. Predict whether the machine is likely to fail.

4. Fashion Outfit Recommendation

Logic

1. Collect user preferences (liked outfits, colors, seasons).
2. Extract outfit features such as color, style, and season.
3. Build a user preference profile.
4. Compare outfit features with the user's profile.
5. Rank outfits based on similarity.
6. Recommend the most similar outfits.

5. Heart Disease Prediction

Logic

1. Load the patient dataset.
2. Clean and preprocess the data.
3. Separate features and target (Heart Disease).
4. Apply feature selection (SelectKBest or RFE).
5. Keep only the most relevant medical features.
6. Train the classification model.
7. Predict heart disease risk.

6. Job Recommendation System

Logic

1. Collect job details and user resumes.
2. Extract skills, experience, and qualifications.
3. Build a user profile.
4. Compare user skills with job requirements.
5. Calculate similarity scores.
6. Recommend the most suitable job listings.

7. Smart Home Energy Prediction

Logic

1. Load the sensor dataset.
2. Clean and preprocess the data.
3. Standardize numerical features.
4. Apply PCA to reduce the number of features.
5. Train the prediction model using the reduced features.
6. Predict future energy usage.

8. E-Learning Tutorial Recommendation

Logic

1. Collect quiz scores and learning history.
2. Identify the learner's weak topics.
3. Match tutorials to those weak topics.
4. Rank tutorials based on relevance.
5. Recommend the best tutorials to improve the learner's skills.

9. Social Media Engagement Prediction

Logic

1. Load the social media dataset.
2. Clean and preprocess the data.
3. Separate input features and target (likes/comments).
4. Apply feature selection to identify important features.
5. Train the prediction model.
6. Predict post engagement.
- 7.

10. Django Attendance App

Logic

1. Create a Django project.
 2. Create a new app named attendance.
 3. Register the app in `INSTALLED_APPS`.
 4. Create models for attendance records.
 5. Run migrations to create database tables.
 6. Create views to handle attendance operations.
 7. Configure URLs.
 8. Create HTML templates.
 9. Run the server and test the attendance application.
- .